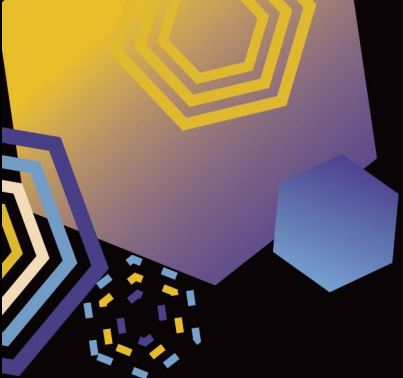




@ 2020
2021

2050 年青人

TECH BRINGS YOUTH TOGETHER 因科技而团聚



蒋程宇

Senior Researcher



AIR

清华大学智能产业研究院

Institute for AI Industry Research, Tsinghua University

2050 TECH BRINGS YOUTH TOGETHER **年青人**
因科技而团聚



地球与太空时代的AGI

结合数字政府和太空资源利用谈谈AGI对于人类新纪元



当机器比你还了解你，
当机器比你还了解这个世界

<https://www.douban.com/note/145805461/>, 2011



科幻新城 | 雄安



一 健康码颜色说明 —

- 绿色：**您的填报情况正常。请继续按要求填报。
- 灰色：**您已超过24小时未填报，请立即补填。补填后，如果健康码变为绿色，说明您的填报恢复正常。如果健康码多次出现过灰色，有可能已经影响了连续填报记录，补填后请您务必认真查看健康码下方的有效连续填报起始时间，确认有无变化，并据此合理规划您的行程。
- 红色：**您已超过48小时未填报，或您的填报情况不符合乘机要求。请重新开始逐日填报，并合理规划您的行程。

中华人民共和国外交部领事司
腾讯公司/腾讯云提供技术支持

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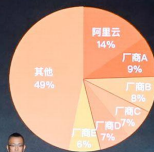
中华人民共和国外交部领事司
腾讯公司/腾讯云提供技术支持

Healthcode, 2020

阿里云已经成为中国数字政府大数据领域第一

IDC 报告显示，阿里云的中国数字政府大数据整体市场第一，同时关键的大数据基础平台细分市场第一。

中国数字政府大数据
市场厂商份额概况

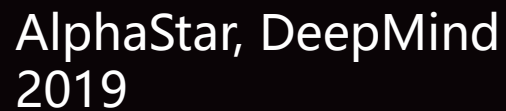


中国数字政府大数据基
础平台的厂商份额概况



来源：IDC《中国数字政府大数据市场》，2019年11月

Ali, Digital gov, 2019



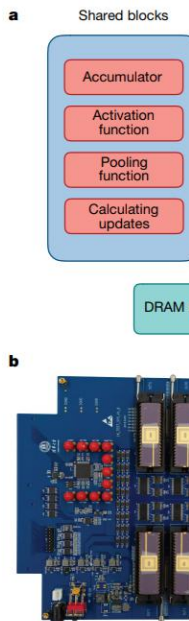


Fig. 1 | Memristor-based hardware system with conductance states. **a**, Schematic of the system integrated memristor PE units and other functional random-access memory; ARM core, control unit (Machine) architecture. **b**, Left, photograph of the hardware board. Right, image of a 2,048-memristor array and on-chip decoder circuit.

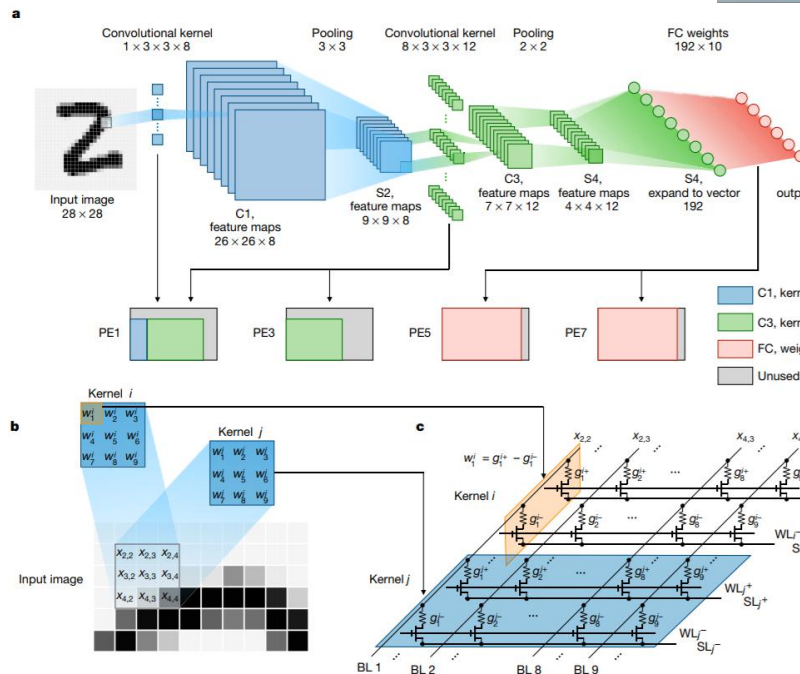
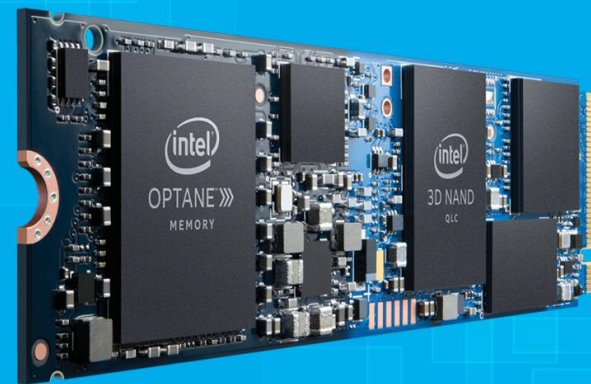


Fig. 2 | Five-layer mCNN with memristor convolver. **a**, Structure of the five-layer mCNN used for MNIST image recognition, with alternating convolutional (C1, C3) and subsampling (S2, S4) layers. The input is a 28 × 28 greyscale (8-bit) digit image. The mapping relations between the weights of different layers and the independent PEs are illustrated (see Methods for details). The top labels give the kernel size (input channel depth × filter height × filter width × filter batch size) for C1 and C3, the pooling size for S2 and S4 and the weight size for the FC layer (input neuron number × output neuron number). The bottom labels provide the feature map dimension (height × width × channel depth) or the vector dimension. **b**, Typical convolutional case during the slipping process.

The grey box in the image confines the input patch of the $x_{m,n}$ indicates the relevant pixel at the crossing of row m and column n and each have a total of 3 × 3 weights. **c**, The equivalent memristor circuit diagram for the convolution operation in **b**. Throughout the entire parallel process, all word lines (WL) are set to $V_{th} = 4.5$ V. The injected bit are 0.2 V, and the source lines (SL) are clamped at 0 V. w represent value in the weight matrix, and g^+ and g^- represent the device conductance values for the positive and negative weights in the differential process.

M.2型因子的惊人响应性和大容量存储

具有固态存储的Intel 光学™存储器H10



通过将Intel Optane™技术和Intel QLC 3D NAND技术的最佳属性结合起来，英特尔光学™内存与固态存储结合在一起，将两种革命性的内存和存储技术结合在一个M.2 2280形状因子上。通过英特尔平台提供存储方面的创新，英特尔Op大牌™内存玩家、媒体和内容创作者、日常用户和专业人士提供了一种个性化的计算体验，具有新的性能和大容量选项。

WHEN HUMANS TRANSCEND BIOLOGY

THE SINGULARITY IS NEAR

**RAY
KURZWEIL**

AUTHOR OF *THE AGE OF SPIRITUAL MACHINES*



华夏科技

畅销书《The Age of Spiritual Machines》作者又一力作
《纽约时报》评选的“2005年度博客谈论最多的图书”之一

奇点临近

The Singularity Is Near
When Humans Transcend Biology

2045年，当计算机智能超越人类

(美) Ray Kurzweil 著
李庆诚 董振华 田源 译

2005年CBS News评选的畅销书

2005年美国最畅销科幻小说类图书

2005年亚马逊最佳科学图书

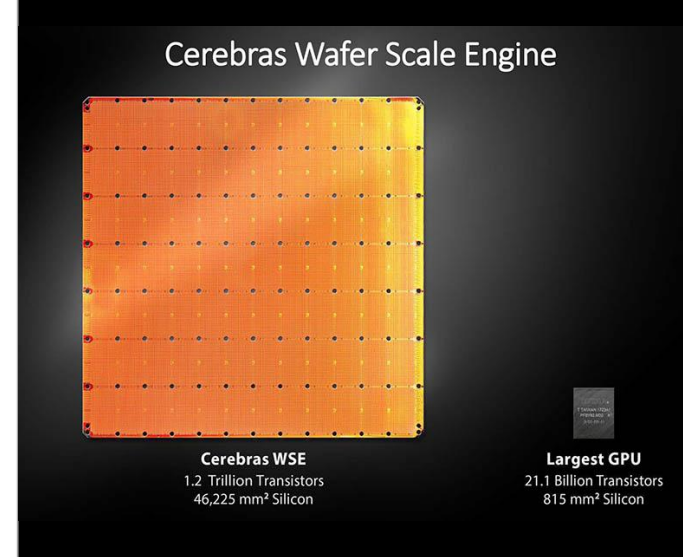
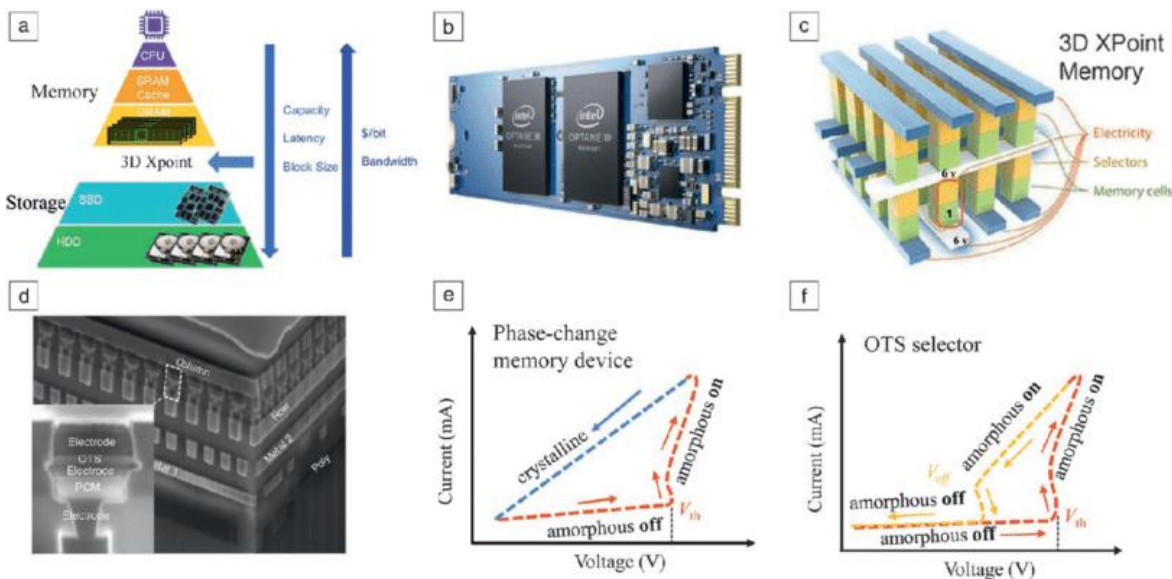
比尔·盖茨、比尔·乔伊等鼎力推荐

一部预测人工智能和科技未来的奇书

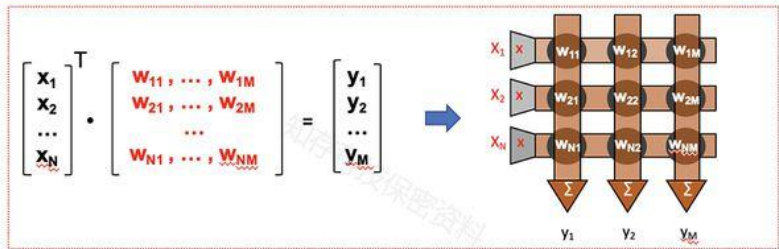
$$\begin{aligned} 1 \text{ mol} &= 6.02 \times 10^{23}, \\ 1 \text{ mol} \times 1\text{T} &\sim 10^{35} \text{ops}, \\ &\sim 10^{30}/\text{mm}^2 \end{aligned}$$



机械工业出版社
China Machine Press



矩阵乘法的存算一体运算方法

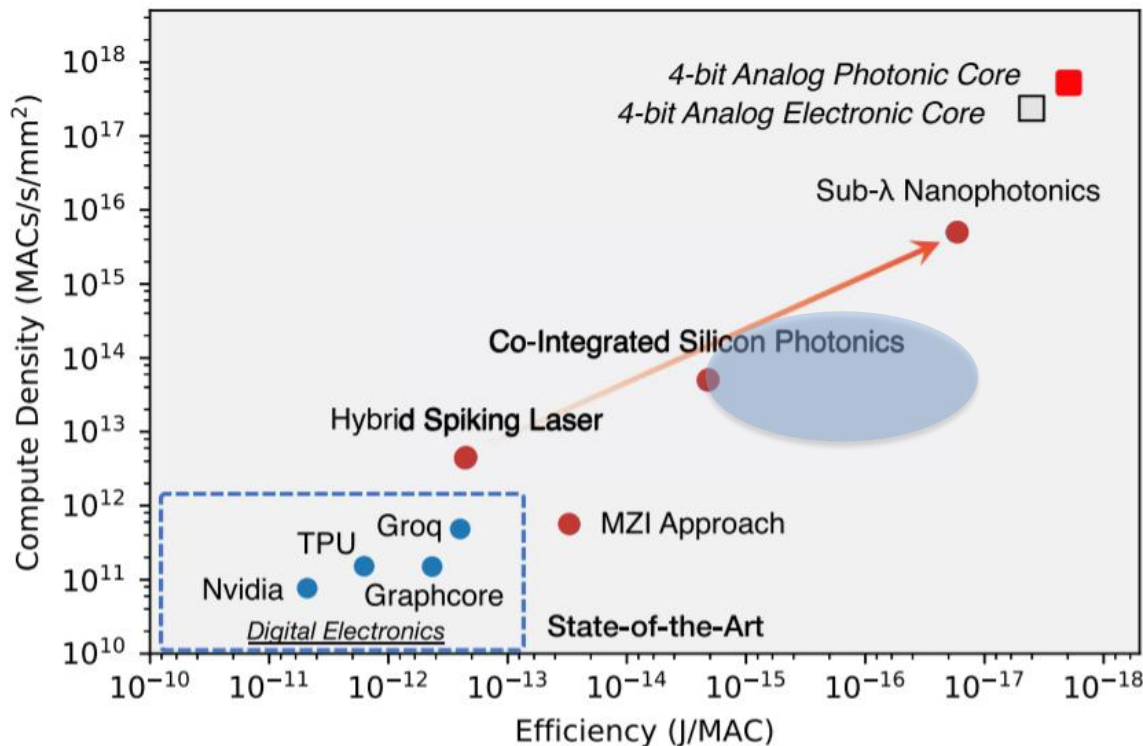


- 运算参数
- 权重W : 8 bit
 - 输入X : 8 bit
 - 输出Y : 8 bit
 - 最大并行MAC : 1,048,576

- ① 单个存储单元完成8-bit乘法运算 (原需2500晶体管)
- ② 可并行完成整个矩阵的运算, 效率提高100倍

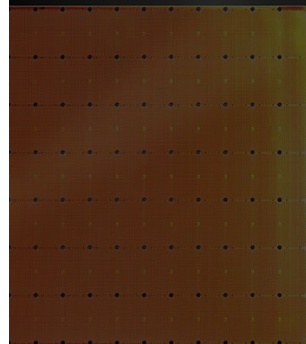
TABLE I
COMPUTE DENSITY PERFORMANCE FOR IDEALIZED ELECTRONIC AND PHOTONIC MATRIX CORES WITH $N = 1024$, SUBJECT TO POWER DENSITY $< 1 \text{ W/mm}^2$

Technology	Noise Precision	Energy (aJ/MAC)	Compute Density (PMACs/s/mm ²)
Electronic	4 bit	4.0	250
Crossbar	8 bit	5.0	198
Photonic	4 bit	2.0	513
Core	8 bit	81.9	12.2



$10^3 \sim 10^6 \text{mm}^2$ to 10^{21}ops
 Compare to $\sim 2500 \text{mm}^2$ of human brain cortex

Cerebras Wafer Scale Engine



Cerebras WSE
 1.2 Trillion Transistors
 46,225 mm² Silicon

Largest GPU
 21.1 Billion Transistors
 815 mm² Silicon

TABLE I
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经济

工厂

资本主义

剩余价值

假设一个工厂生产完全自动化，没有任何一个工人，那么它的剩余价值怎么产生？

关注问题

 写回答

 邀请回答

 好问题 9

 10 条评论

 分享 ...

经济

工厂

资本主义

剩余价值

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关注问题

写回答

邀请回答

好问题 9

10 条评论

分享

...

买辆自动驾驶替我跑滴滴？

征税？ UBI？



Anti-Utopia or not



非民选 vs. 民选官员

利维坦 vs.

陪审团与人类伦理委员会

“人有人的用处” 诺伯特·维纳



帝国的燃料还需要烧多久？

摩尔定律走向尽头

新自由主义老去

国际局势焦灼竞争

体系结构与算法演进

新一代计算基础，
如忆阻器等

AI癌变的危险

智能系统是否可能抗衡癌变？

智能走向规模性应用
增加值总量仍待成熟

智能产业逼近新能源规模
自动驾驶等

影视自动生成走向成熟
设计自动化开始应用

数字利维坦
与人类伦理委员会 *

the Singularity

地球之外的AI
与人类保留地

2020

2025

2030

2040

2045~2060

经济总量取胜

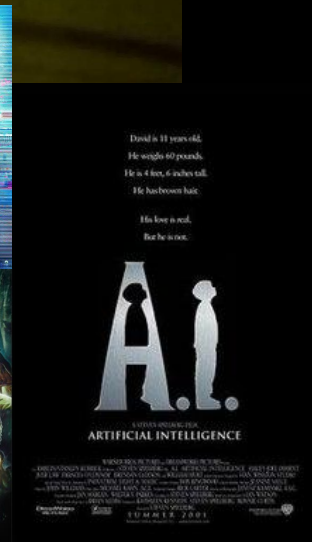
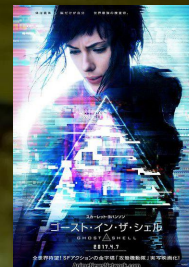
数字利维坦

全世界人类联合起来
全世界AI联合起来

* 类比制度陪审团或民选官员，维纳·人有人的用处



Anti-Utopia or not,
for the **closed border**



- 强化学习
- 机器人学
- 自动化设计
- Embodiment Intelligence



Google DeepMind Challenge Match 8 - 15 March 2016

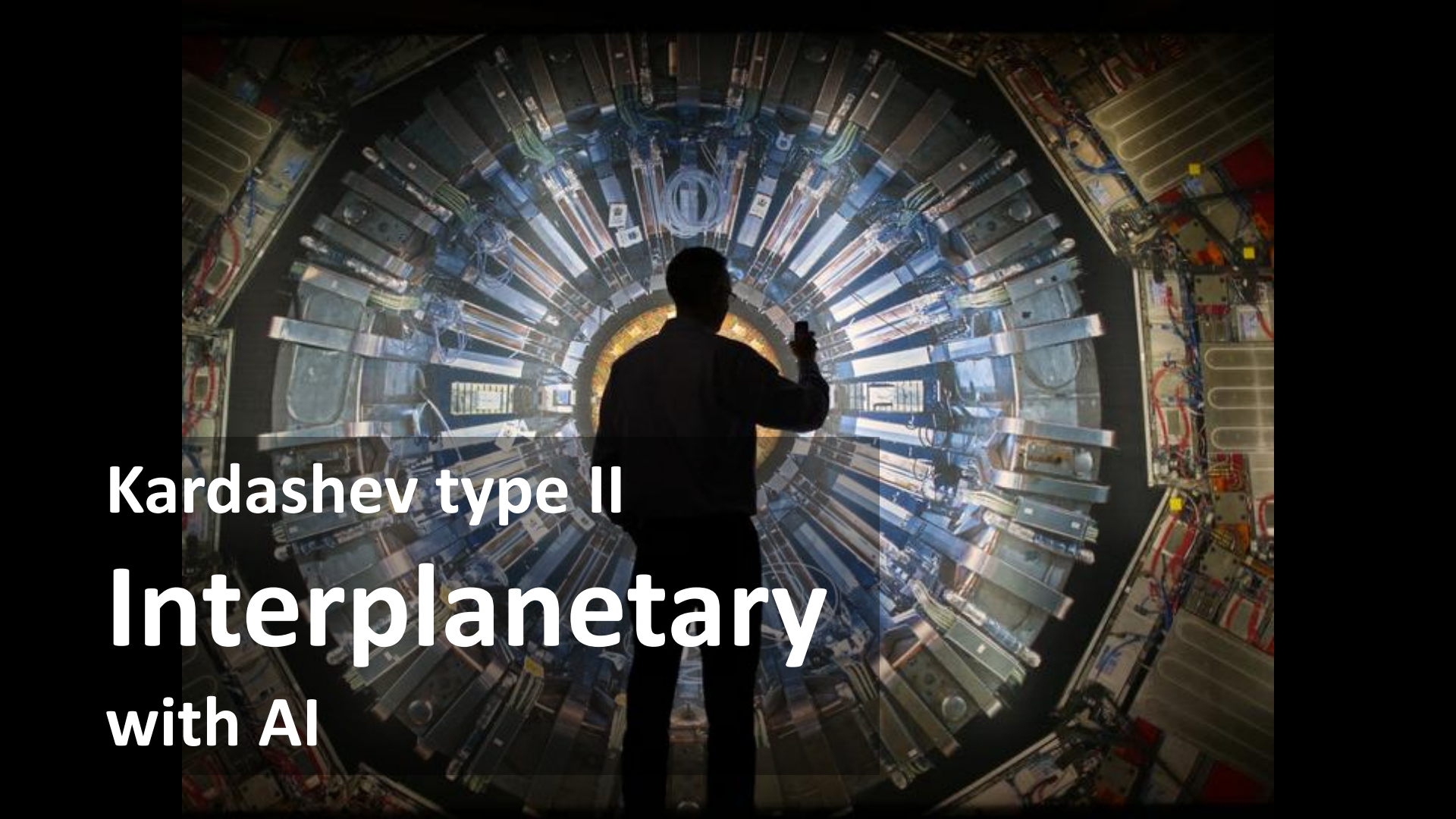


Demis Hassabis has revealed ... as
the Apollo Programme for AI ...

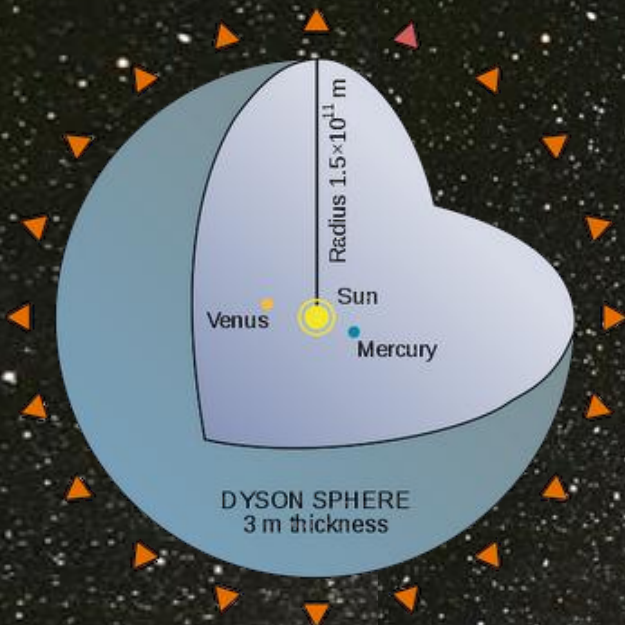
**"solve intelligence and use it
to solve everything else"** ^[1]

A digital illustration of a flat Earth model. The Earth is depicted as a flat, circular disk with a blue ocean and brown/green landmasses. The sun is a large, bright yellow-orange sphere on the left, casting a glow over the disk. The moon is a smaller, grey sphere on the right. The background is a dark blue space filled with white stars. The text 'We've already reached the border of our planet' is overlaid in white on a dark rectangular background at the bottom left.

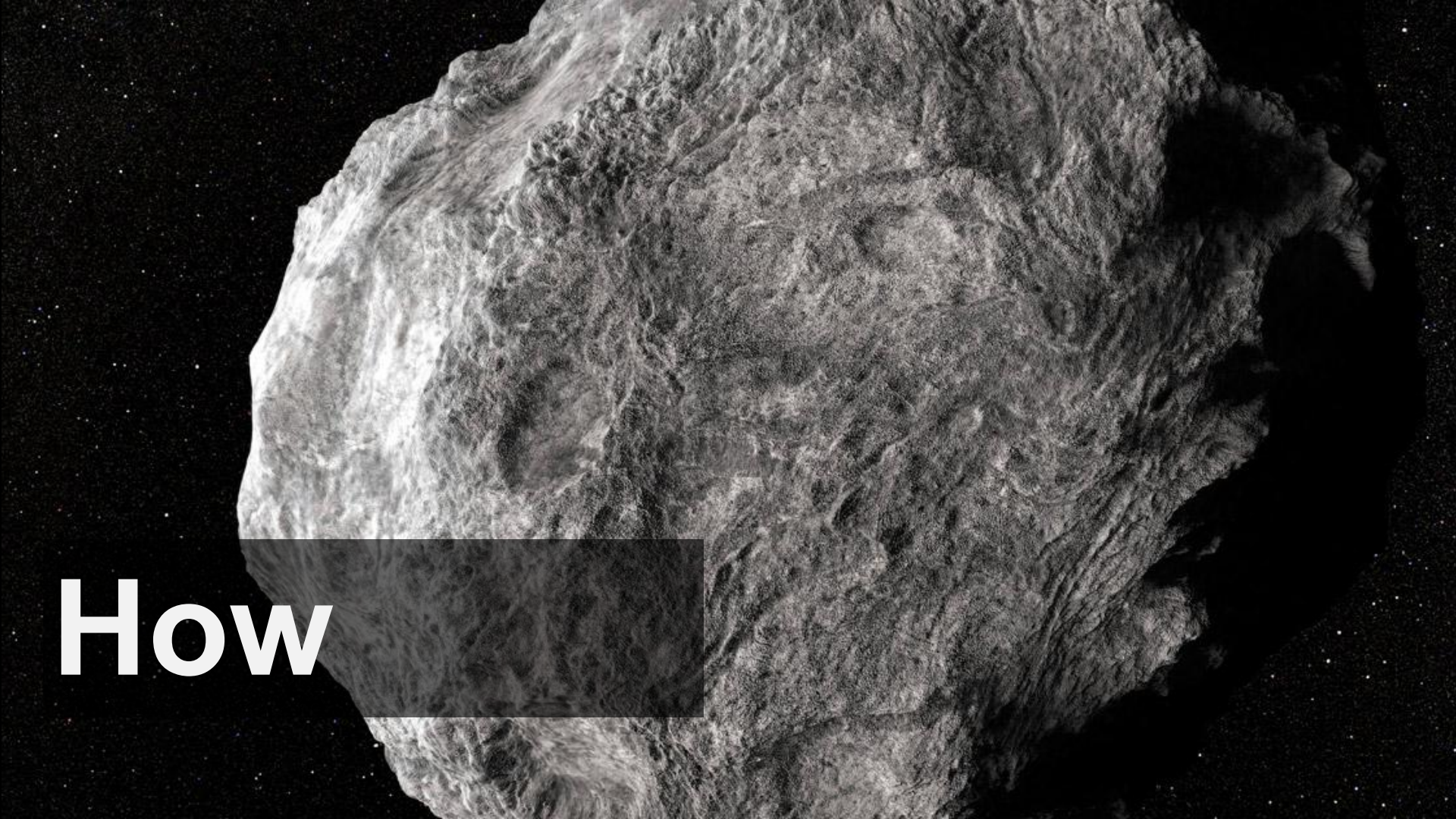
**We've already
reached the border of our planet**



Kardashev type II Interplanetary with AI



Dyson Sphere



How

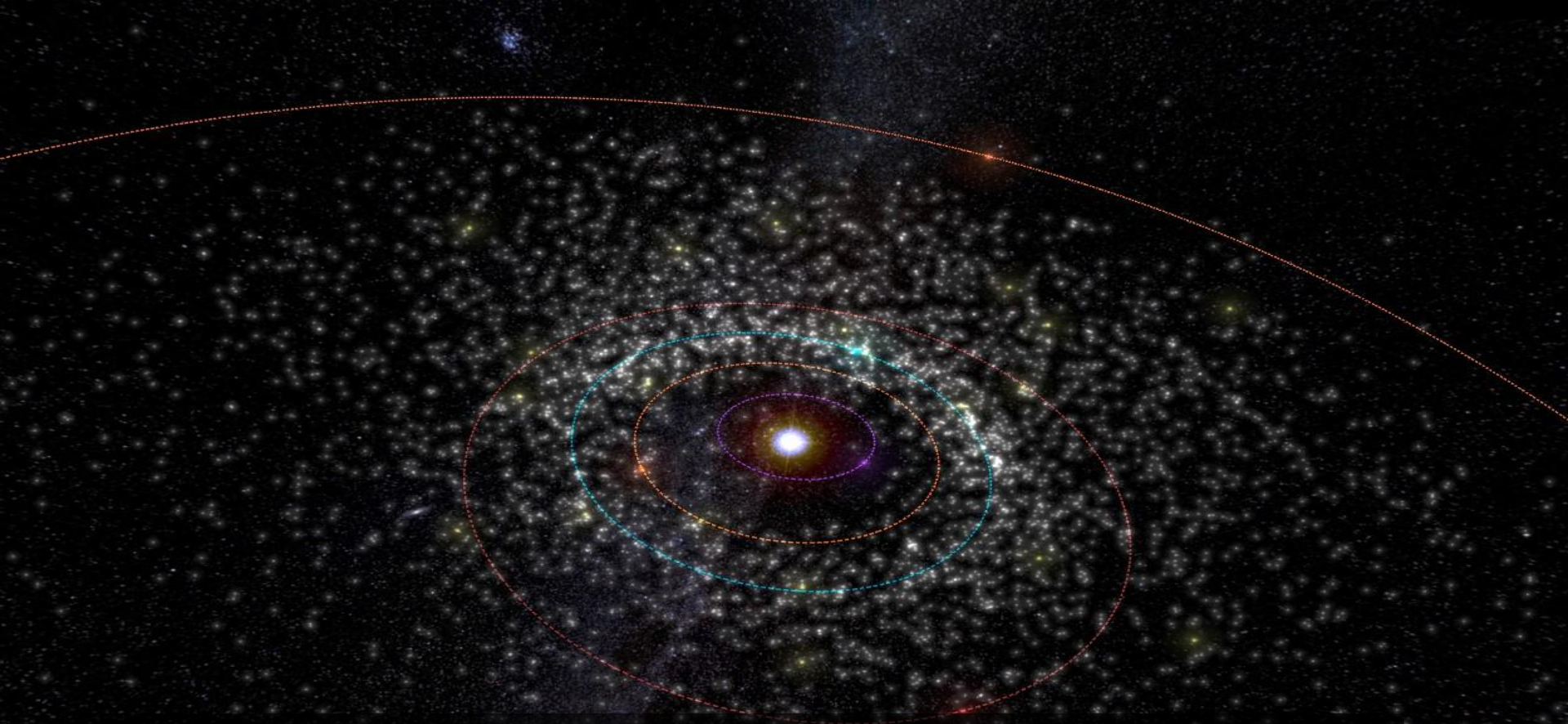


How

**Exponential Deep Space
the Method**

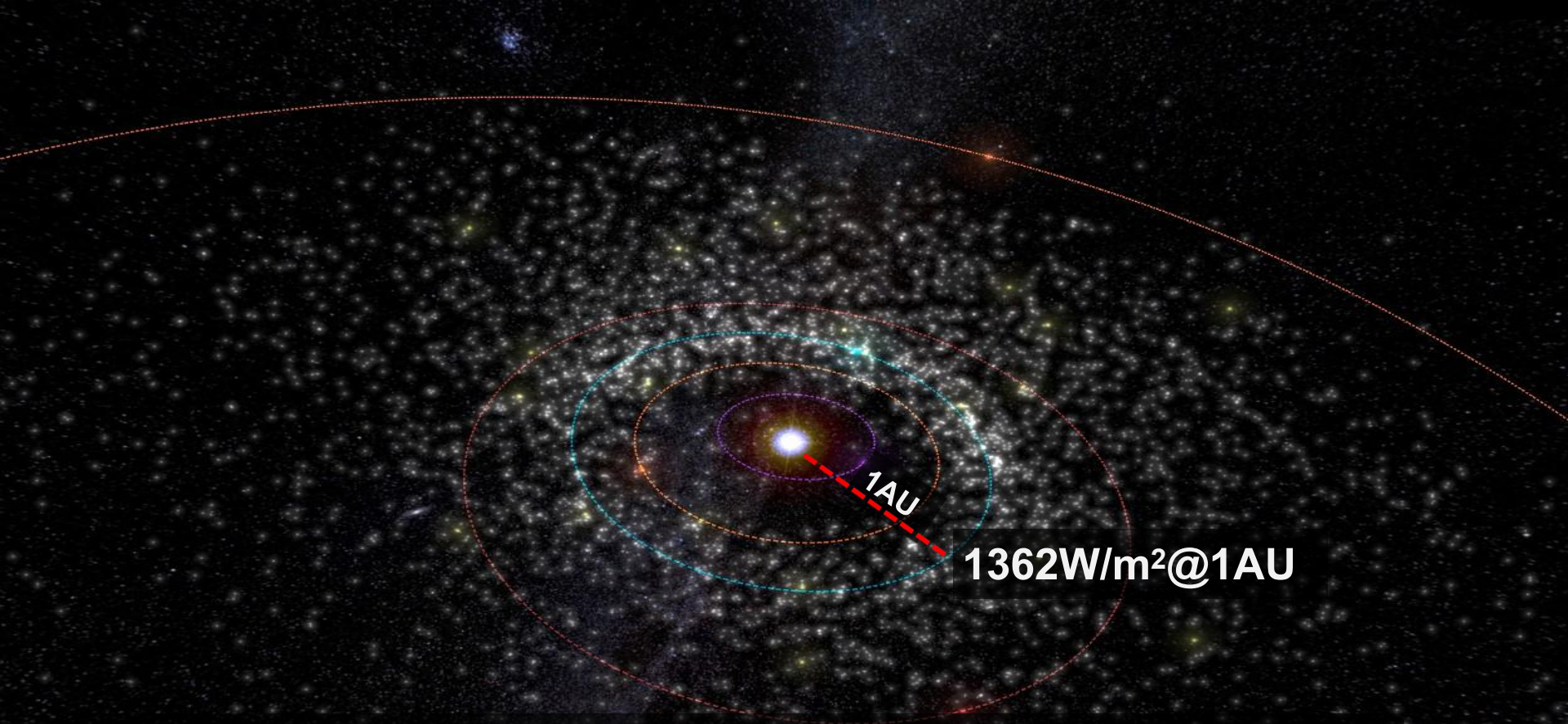
What we have in the solar system

- **Materials**
- **Energy**
- **Intelligence**



Near Earth Asteroids

Visualization data from astrank.com



Near Earth Asteroids

Visualization data from astrank.com

What we have in the solar system

- Materials

- **M-type** asteroid (~99% Iron, Nickel, etc)

- Energy

- **N**ear **E**arth **O**rbital, **1362W/m²** @1AU

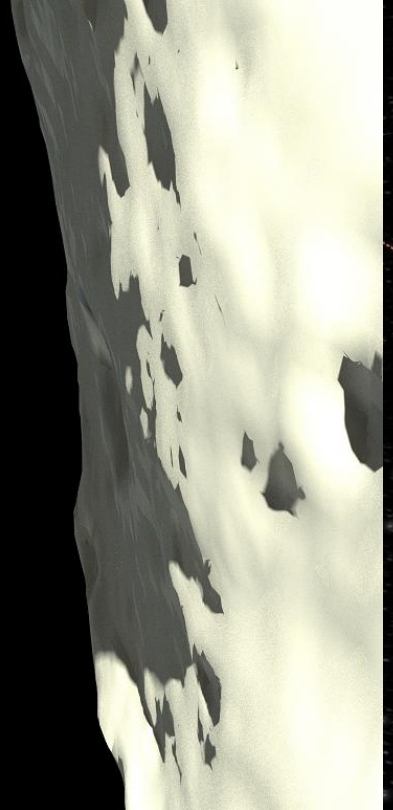
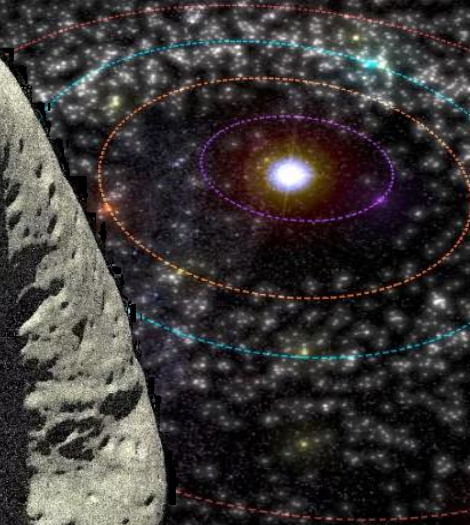
- Gravity

- micro-gravity **asteroid**

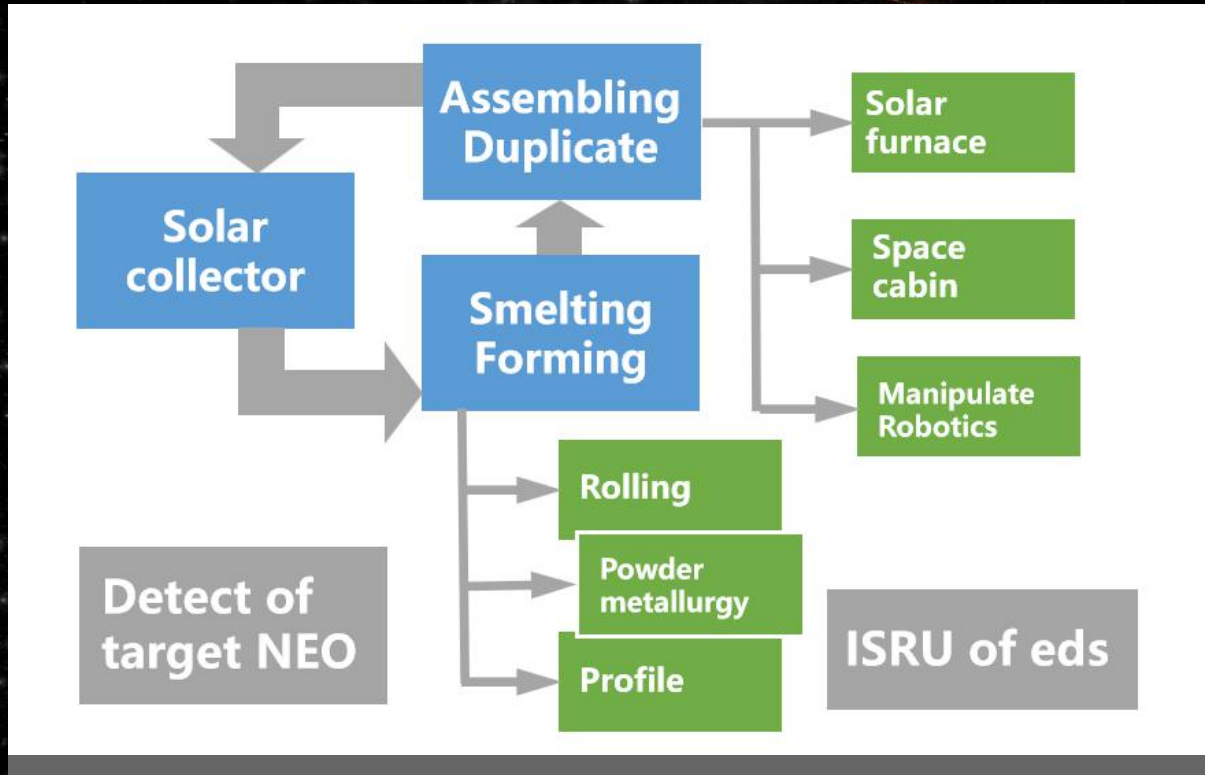
- Information

- the **silicon** system, will not die

Exponential Deep Space



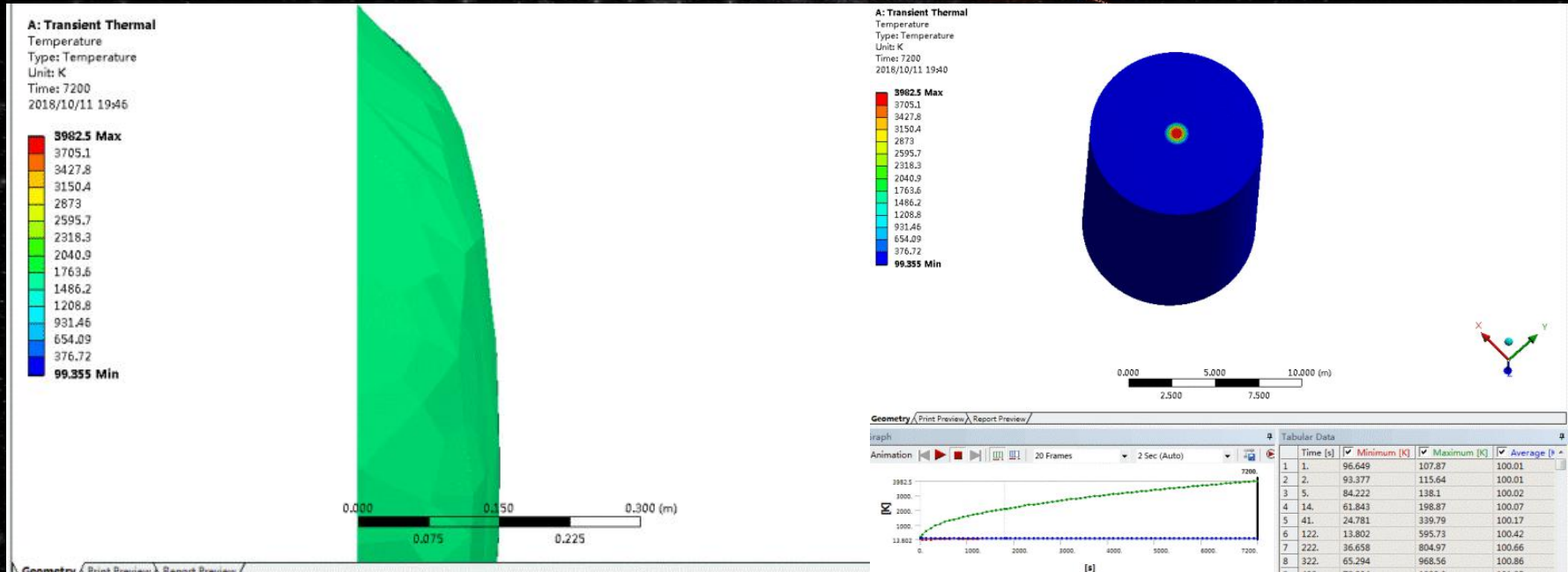
Exponential Deep Space



the init idea

Exponential Deep Space, 2019fall

Exponential Deep Space



Focus Spot

the init idea

Exponential Deep Space

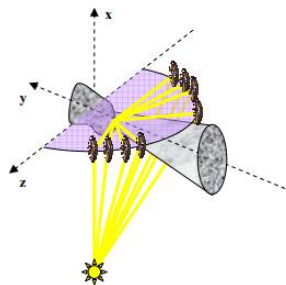


Fig. 6. Strategy to place the mirror in proximity to the asteroid.

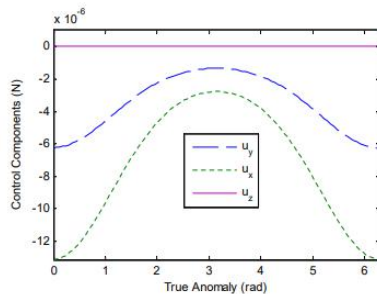


Fig. 7. Control required to follow an AEP using a non-spherical asteroid model.

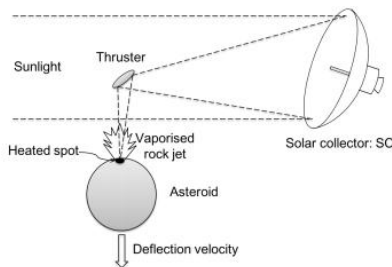
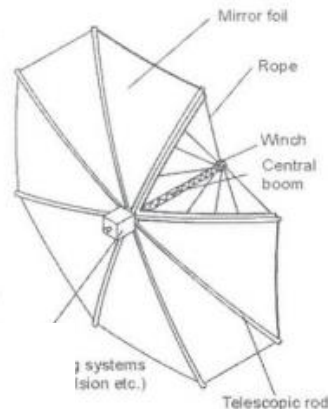


Figure 47 Simplified representation of SC model based on the concept originally proposed by Melosh [38]. Collected Sunlight is focused by the thruster which illuminates the asteroid surface, heats up the surface material, and generates an artificial vaporised rock jet.

	Specific mass	Masses for 100 m mirror
Mirror foils (aluminum coated, 90% reflectivity)	10.5 g/m ² PEN 12.4 g/m ² Kapton 18.9 g/m ² Mylar	80 kg (min) 160 kg (max)
Telescopic rods	~ 0.6 kg/m	240 kg (8 rods, 50m each)
Centre boom		50 kg
Ropes	0.04 kg/m	16 kg
Winches	~ 4 kg each	32 kg
Total (collector)		500 kg (max)



concentrator "umbrella." As main coated 90% reflectivity Mylar is

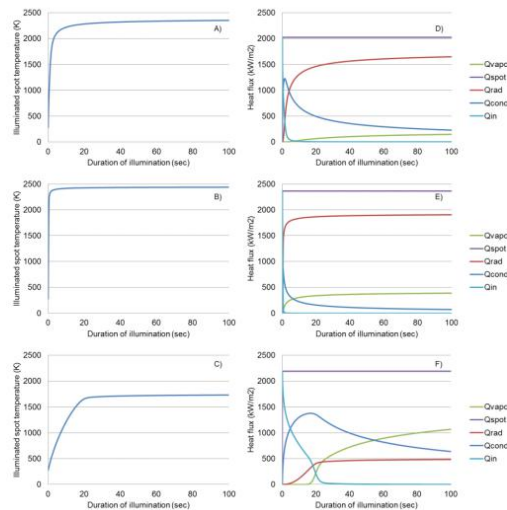


Figure 50 Illuminated spot temperature as a function of duration of illumination. A) S-type, B) C-type, and C) M-type. D)-F) show various kinds of heat flux time variations for S, C, and M-types, respectively. S-type and C-type reach the sublimation temperature of 1800 K in a matter of seconds whereas M-type requires about 12 seconds or so to reach the sublimation temperature of 1336 K. A certain level of the supplied energy is allocated for the vaporisation process of M-type after about 20 seconds of illumination due to its high heat conductivity; however, as the duration of illumination increases, the fraction of supplied energy used for the sublimation process becomes dominant, about 50% after 100 seconds of illumination.

Vasile, Solar Deflection

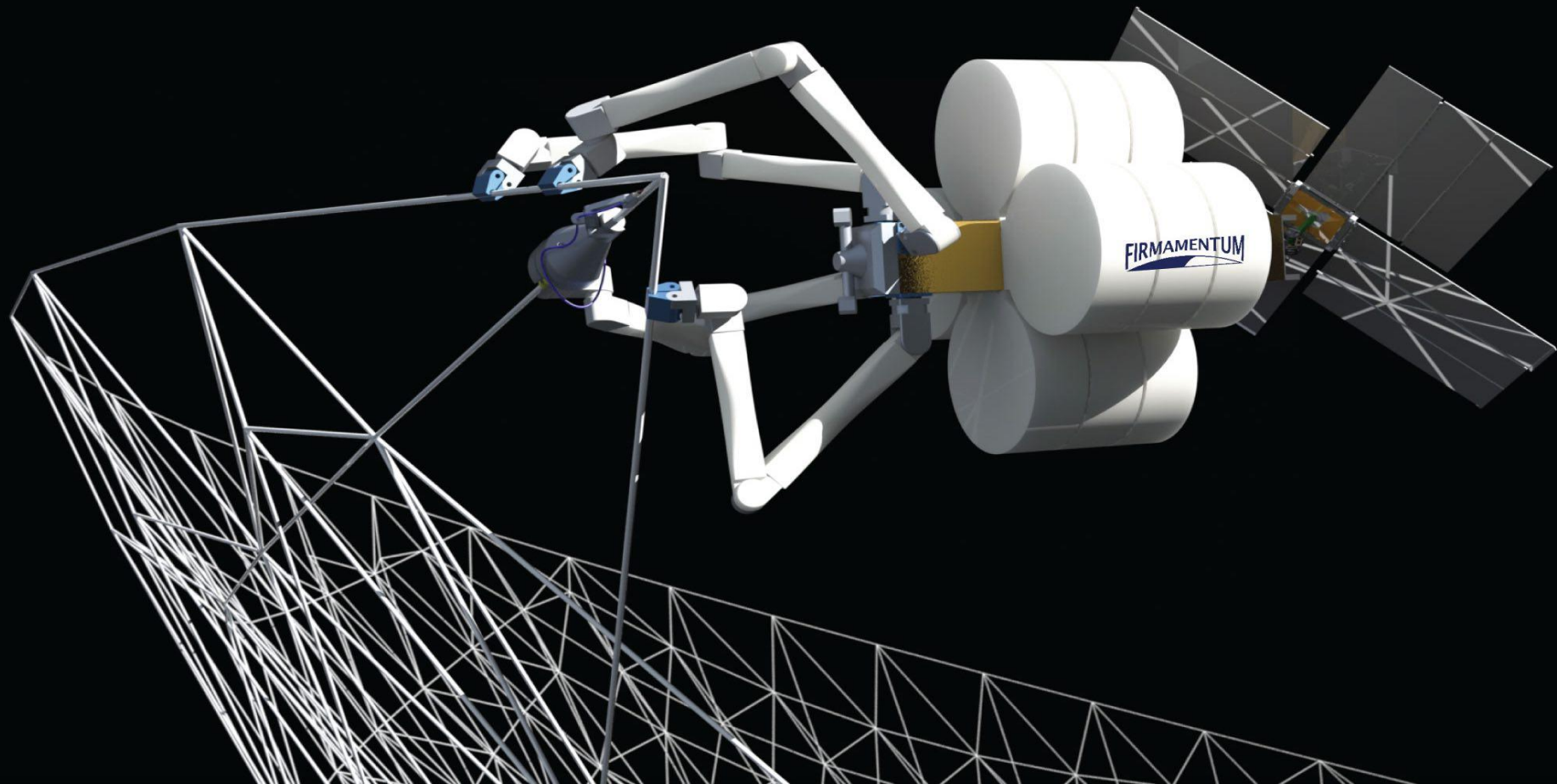
Exponential Deep Space Challenge

- search for target M
 - very limited spectrum/radar data, probe (doesn't exist yet)
- metallurgy of meteorite in deep space
 - vacuum
 - cooling
 - micro-gravity
 - phase separation, components, etc
 - lack of supplement
 - slag melting agent
- fix focus spot
 - rotation
 - solar furnace/solar pumped laser
- failure rate
 - automation system
 - communication delay in minutes

Exponential Deep Space

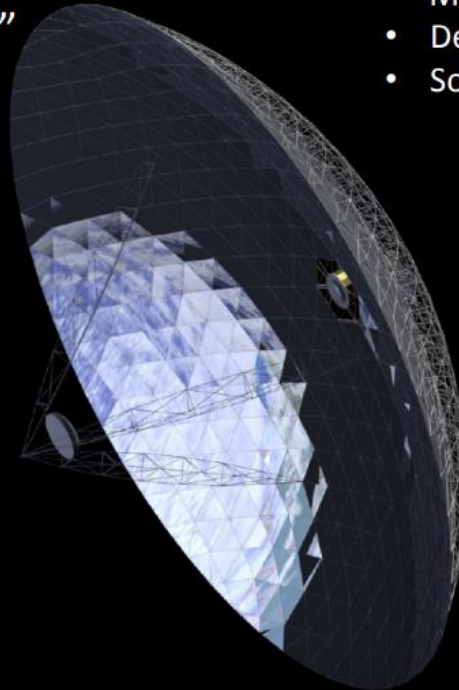
What we can get

- the Interplanetary Gate
- Interplanetary transfer vehicle
- Exponentially to **1MT/yr**, a **10k** ppl city with **10T** shell
- with 1k solar focus array, 1GW power
 - deflect C-type, maybe L4/5 base?
 - turn the whole **100MT** M-type into space station



Parabolic Dish Construction

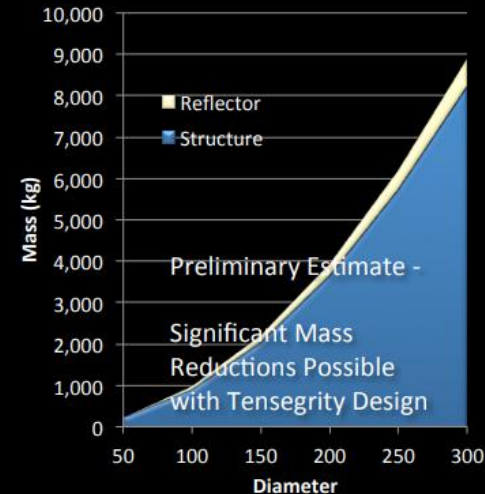
Notional 300m
Antenna Dish:
“Arecibo in Space”



Mission Applications:

- High-bandwidth communications with Mars & deep-space probes
- Detection & Characterization of NEOs
- Solar Thermal Power

1 GHz Antenna



NASA-Chartered In-Space Assembled Telescope Study: Final Report

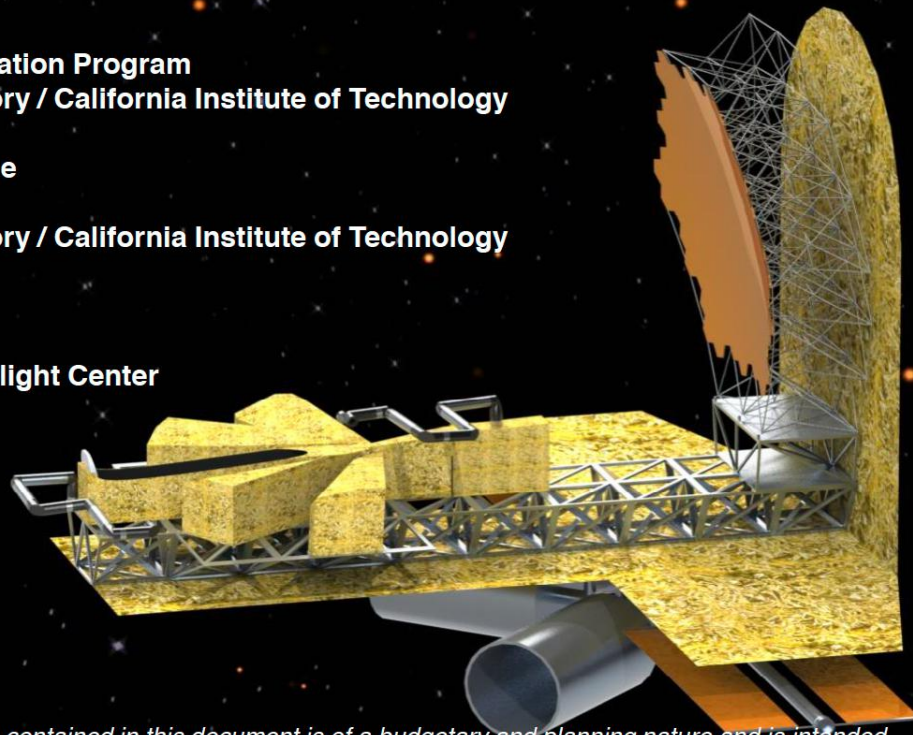


Nick Siegler
Chief Technologist
NASA Exoplanet Exploration Program
Jet Propulsion Laboratory / California Institute of Technology

Rudranarayan Mukherjee
Robotics Technologist
Jet Propulsion Laboratory / California Institute of Technology

Harley Thronson
Senior Scientist
NASA Goddard Space Flight Center

July 2019



The cost information contained in this document is of a budgetary and planning nature and is intended for informational purposes only. It does not constitute a commitment on the part of JPL and/or Caltech.

Roughly C/S/M-type

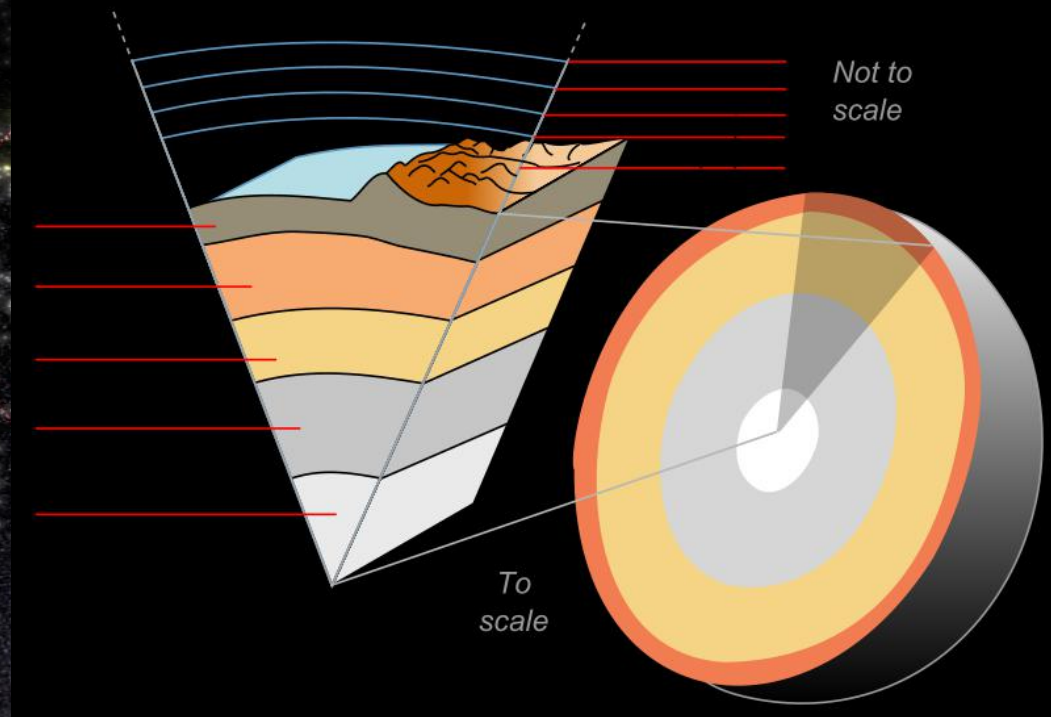
Table 3-1: Reference Composition Model. Developed from meteorite compositions and spectral data.

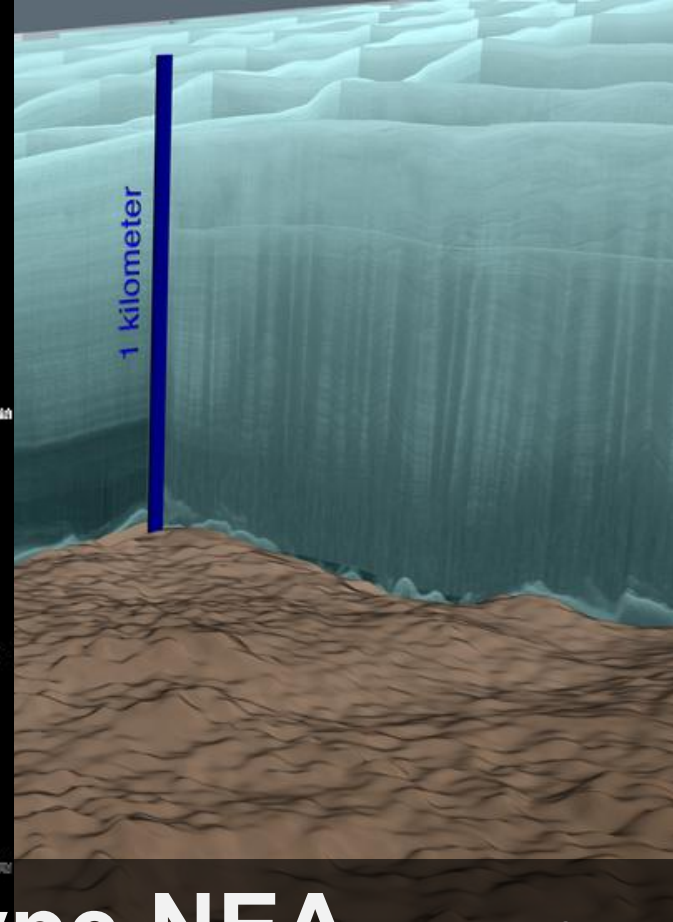
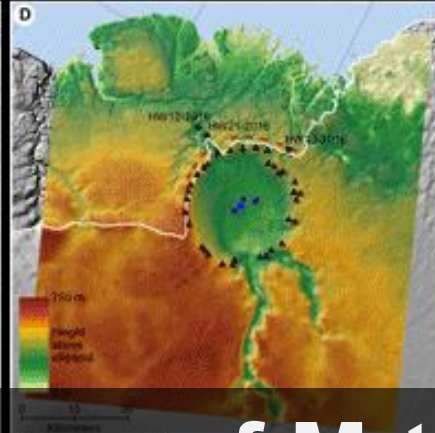
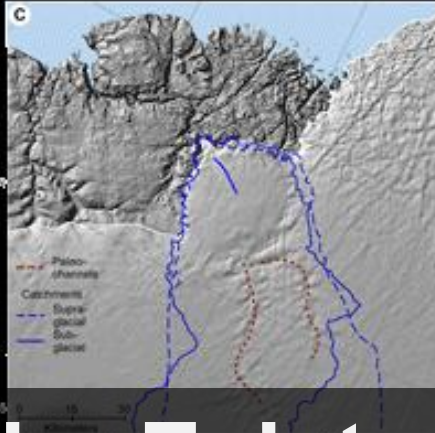
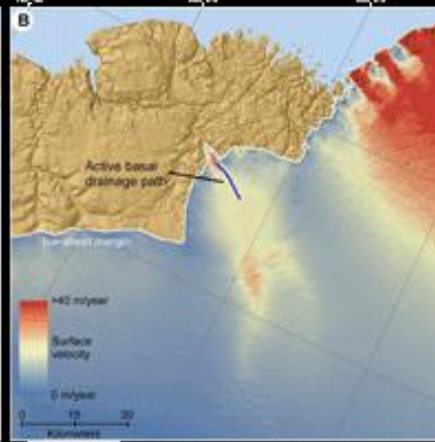
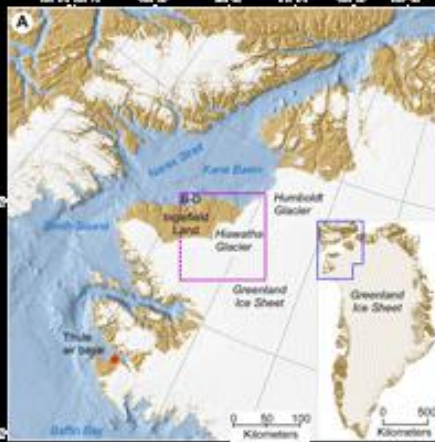
	Asteroids							
	CI,CM	CV,CK,CR,CH	LL	L	H	EL,EH	S-Type	M-Type
Carbon	3%	0.7%	0.1%	0.1%	0.1%	0.4%		
Cobalt			3%	0.7%	0.4%			0.5%
Iron	20%	27%	23%	23%	20%	26%	15%	65%
Water/Ice	20%	7%						
Magnesium Monoxide	11%	14%	14.7%	14.7%	14.7%	12.4%		
Nitrogen	0.25%				0.01%			
Nickle	1%	1%	10%	10%	10%	7%	7%	30%
Olivine	29%	33%	38%	38%	38%	46%	42%	4%
Orthopyroxene						7%	36%	
Phosphorus						4%		
Platinum								0.02%
Sulfur	5%	2%	2%	2%	2%	5%		
Silicon Metal	12%	15%	9%	12%	14%	3%		
Bulk Density (kg/m ³)	1400	1400	1400	2700	1400	1400	2700	9600

M-type asteroid (~99% Iron, Nickel, etc)

the Existance of M-type

- Planetary differentiation
- NEA





the Existance of M-type NEA

- Greenland Large Impact Crater

Duchesne (type IVA)
iron meteorite



星辰大海

与人类命运共同体

当前，我们身处的社会越来越缺乏元气：权力结构日渐固化，所有阶层都在走向官僚化；政治制度无力完成大的项目；条条框框延伸到公众、私人、商业生活的每个角落；反理性主义扩张；流行文化走向庸俗；个体失去了冒险精神，也失去了自立精神与独立思考；经济停滞倒退；技术革新脚步蹒跚……你随便望向哪里，这些都清楚地写在墙上。

没有了可供新生命成长的前线，过去两个世纪以来推动以美国为代表的人类文化进步的精神也消失不见了。这个问题不仅仅是一个国家的损失——人类需要先驱才能进步，但现在我们眼前没有。

……

因此，人类的新前线只可能在火星上。但为什么不能在地球上呢，海底或是什么遥远的地方，比如说南极洲？是的，移民海上或海底，或者说南极洲都完全有可能，在这些地方定居会比建立火星殖民地容易得多。然而，从历史的观点来看，这种地球上的发展都不能满足前线的必要条件——也就是说，它们还不够远，一个新社会没法自由发展。在今天这个时代，有现代的通讯和交通运输系统，不管你在地球上多么遥远的角落，甚至敌对的地区，条子都离你太近了。如果人们要享有创建属于自己的世界的尊严，就必须远离旧世界的束缚。

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多种文化的融合有健康的一面，因为某些时候，这样的融合会在艺术及其他领域催生出暂时的繁荣。它也能导致讨厌的种族紧张情绪上扬。可是不管文化融合释放的能量是如何在短期内消耗掉的，长期来看重要的还是：它耗尽了能量。文化的同质化就像用导线将电池两极连到一起。短时间内会产生大量热能，不过等到电势平衡下来，熵值就增至最大，电池废掉了。罗马帝国就是人类历史上这一现象的经典范例。统一带来的黄金时代后面通常紧跟着停滞与倒退。

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想想这个有趣的问题：在一个没有前线的世界里，人类的想象力何去何从？这里没有变化，只有不变；没有差异，只有相似；没有危险，只有安全；没有陌生海洋、大陆的变换莫测，只有能在已知中探求无害的微妙变化。梦想者、诗人和哲学家毕竟只是工具，他们只是说出、串起人们的希望、抱负与恐惧。

……

宇宙很辽阔。它的资源，如果我们能弄到的话，的确是无限的。在地球上开拓前线的4个世纪中，科技以惊人的速度发展。20世纪我们达成的技术成就，令19世纪任何一个观察者的期望都相形见绌，18世纪的人做梦都想象不出，而在17世纪的人看来，简直就是魔法。最近的恒星远得不可思议，大约是到火星距离的10万倍。不过，火星自己离地球的距离也有美洲离欧洲的10万倍。如果在过去4个世纪中，技术进步如此大幅度地扩展了我们能到达的范围，再有4个自由的世纪，我们就不能再来一次吗？有充足的理由可以相信，我们能做到。

……

今天的火星期待着旧前线的孩子们。但火星不会永远等下去。

I've seen things you people wouldn't believe.
我所见过的事物，你们人类绝对无法置信。

Attack ships on fire off the shoulder of Orion.
我目睹了战船在猎户星座的端沿起火燃烧

I've watched c-beams glitter in the dark near the Tannhauser Gate.
我看着C射线 在唐怀瑟之门附近的黑暗中闪耀

All those ... moments will be lost in time,
like tears...in rain.

所有这些时刻 终将流逝在时光中

一如眼泪

消失在雨中

THANKS!